Note 1: Weekly note

Author: JiaHao Liang *Date: 2026.4.5*

1.1 Summary of Progress

This weekly note organizes learning insights and rigorous derivations based on the classic textbook *Quantum Computation and Quantum Information* (Nielsen & Chuang):

1. Restructured the core axiomatic system of quantum mechanics in the textbook, refining the mathematical formulation of pure states, observables, measurements and unitary evolution.
2. Supplemented complete formal proofs for key conclusions (density operators, composite systems, pure-state criteria) omitted in the original text.
3. Consolidated core content including basic quantum circuits, representative algorithms and fundamental information theory in the textbook.
4. Reinforced the mathematical foundation with Lie group-related derivations, deepening the understanding of unitary evolution and gate decomposition.

All content integrates textbook knowledge and self-organized reasoning, forming a rigorous, systematic reading record.

1.2 Introduction and overview

1.2.1 Timeline Summary of the Development of Quantum Computation and Quantum Information

1. Early Conceptual Period (Early 20th Century Early 1980s)

- 1900s1920s: Modern quantum mechanics was established.
- 1948: Claude Shannon founded classical information theory.
- Late 1960s: Stephen Wiesner proposed early quantum cryptographic ideas.
- 1982: Richard Feynman pointed out classical computers cannot efficiently simulate quantum systems.

2. Theoretical Foundation (Mid1980s Early 1990s)

- 1984: Bennett and Brassard proposed the BB84 protocol.
- 1985: David Deutsch introduced the universal quantum Turing machine.
- 1992: Superdense coding developed; no-cloning theorem formalized.

3. Breakthrough Era (MidLate 1990s)

- 1994: Peter Shor invented Shors factoring algorithm.
- 1995: Lov Grover developed Grovers search algorithm.
- 1996: CSS codes and stabilizer codes discovered.

4. Formation of a Discipline (Around 2000)

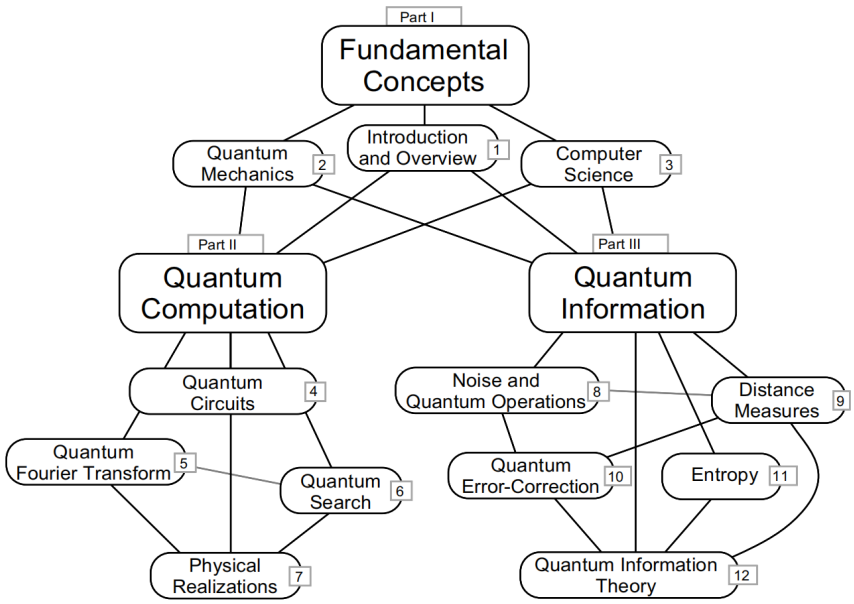


Figure 1. Structure of the book.

- Quantum teleportation experimentally demonstrated.
- Subfields including distributed quantum computation emerged.
- Publication of Nielsen & Chuangs textbook marked the fields establishment.

Logical Flow: physical and information-theoretic preliminaries $\rightarrow$ formal computational models $\rightarrow$ proof of quantum speedup $\rightarrow$ solutions to noise and errors $\rightarrow$ formation of a complete scientific discipline.

1.2.2 Single-Qubit State : Preparation and Measurement with Polarized Light

We build and measure an arbitrary single qubit using only polarized light and optical components, with full physical intuition.

1. Arbitrary Single-Qubit State Preparation

- Optical path: Light source $\rightarrow$ polarizer $\rightarrow$ half-wave plate $\rightarrow$ quarter-wave plate
- Polarizer: Produces pure vertical $|0\rangle$ or horizontal $|1\rangle$
- **Half-wave plate** (rotated by θ): Generates arbitrary linear polarization

$$|\psi\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle$$

- **Quarter-wave plate** (introduces phase ϕ): Creates the full complex qubit state

$$\boxed{|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle}$$

2. Quantum Measurement Rule

Choose an orthogonal basis $\{|a\rangle, |b\rangle\}$, photon() collapses with probabilities

$$|\langle a|\psi\rangle|^2, \quad |\langle b|\psi\rangle|^2$$

In polarization optics: measurement = polarizer + single-photon detector

3. Two Standard Measurements

- Measurement in $\{|0\rangle, |1\rangle\}$ (computational basis): Place a vertical polarizer followed by a detector.

$$P(0) = |\langle 0|\psi\rangle|^2, \quad P(1) = |\langle 1|\psi\rangle|^2$$

- Measurement in $\{|+\rangle, |-\rangle\}$ (diagonal basis): Place a 45° polarizer followed by a detector.

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

4. Full Experimental Sequence

- Optical path: Laser $\to$ polarizer $\to$ half-wave plate $\to$ quarter-wave plate $\to$ measurement polarizer $\to$ detector
- State preparation: control amplitude and phase

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

- Measurement: choose basis with a polarizer
- Statistics: repeat many times to obtain

$$P(0) = |\alpha|^2, \quad P(1) = |\beta|^2$$

1.2.3 Quantum gates and circuits

Definition 1 A **quantum gate** is a *unitary linear operator* acting on a finite-dimensional Hilbert space of quantum states. For an n-qubit system, a gate is a unitary operator $U \in \mathrm{U}(2^n)$.

Common single-qubit gates (matrix form and rules)

- Pauli gate (quantum NOT):

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad X|0\rangle = |1\rangle,\ X|1\rangle = |0\rangle$$

- Pauli-Z gate (phase flip):

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad Z|0\rangle = |0\rangle,\ Z|1\rangle = -|1\rangle$$

- Hadamard gate (superposition):

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H|0\rangle = |+\rangle,\ H|1\rangle = |-\rangle$$

- Phase gate :

$$P_\theta = \begin{pmatrix} 1 & 0 \\ 0 & e^{\mathrm{i}\theta} \end{pmatrix}$$

- The Puali matrix

$$X = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

  Define Rotation gate

$$R_x(\theta) = e^{-i\theta X/2}, R_y(\theta) = e^{-i\theta Y/2}, R_z(\theta) = e^{-i\theta Z/2}.$$

Theorem 2 *Let U be a quantum gate with $U^2 = I$, then we have*

$$e^{i\theta U} = (\cos\theta)\, I + (i\sin\theta)\, U, \quad e^{\theta U} = (\cosh\theta)\, I + (\sinh\theta)\, U$$

$$e^{\gamma I + i\theta U} = e^{\gamma}\left(\cos\theta I + i\sin\theta U\right), \quad e^{\frac{i\pi}{2}(I-U)} = U.$$

Proof

$$e^{i\theta U} = \sum_{k=0}^{\infty} \frac{(i\theta U)^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{(i\theta U)^{2k+1}}{(2k+1)!} = \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k} I}{(2k)!} + iU \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k+1}}{(2k+1)!} = (\cos\theta)\, I + i\,(\sin\theta)\, U$$

$$e^{\theta U} = \sum_{k=0}^{\infty} \frac{(\theta U)^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{(\theta U)^{2k+1}}{(2k+1)!} = \sum_{k=0}^{\infty} \frac{\theta^{2k} I}{(2k)!} + \sum_{k=0}^{\infty} \frac{\theta^{2k+1}}{(2k+1)!} U = (\cosh\theta)\, I + (\sinh\theta)\, U$$

■

Common two-qubit gate Controlled-NOT (CNOT) gate: control qubit flips the target qubit if control is $|1\rangle$:

$$\mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \mathrm{CNOT}|xy\rangle = \begin{cases} |xy\rangle & \text{if } x = 0 \\ |x\,(1-y)\rangle & \text{if } x = 1 \end{cases}$$

Definition 3 (quantum circuit) A **quantum circuit** is a finite sequence of quantum gates arranged on ordered qubit wires, starting from an initial product state.

The mathematics form of quantum circuit is product $U = G_1 G_2 \cdots G_n$, where each G_j are quantum gates.

Gates act sequentially or in parallel on selected qubits; the overall operator equals the product of individual unitary gates in time order. Measurements can be appended at the output to collapse quantum states into classical outcomes.

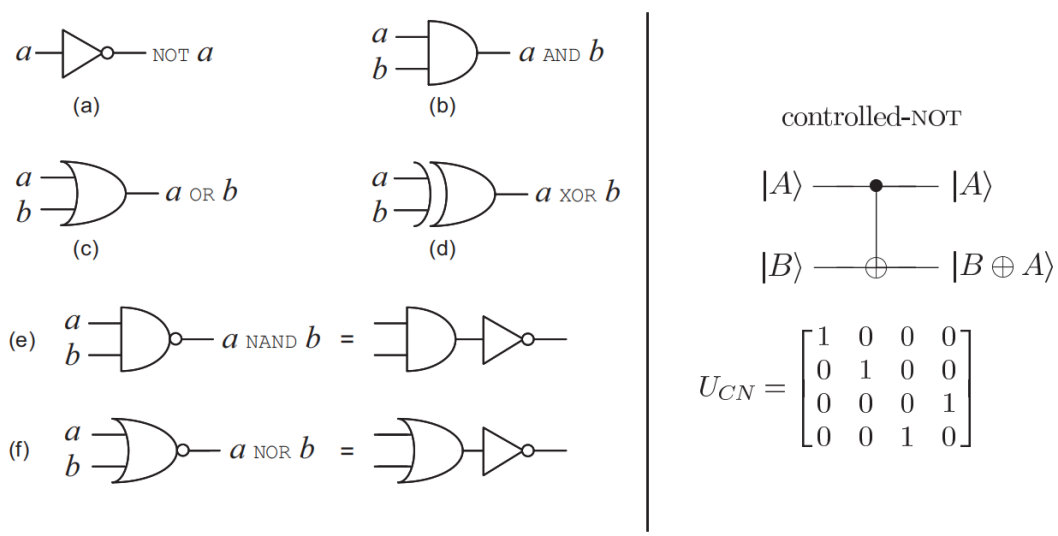
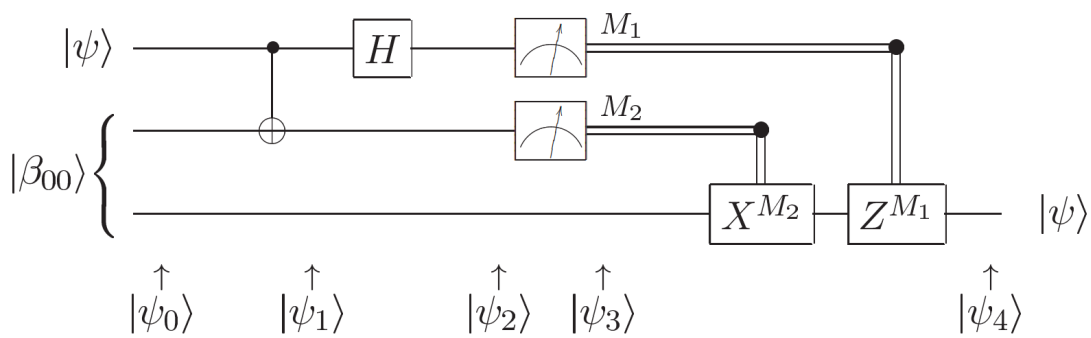


Figure 1.6. On the left are some standard single and multiple bit gates, while on the right is the prototypical multiple qubit gate, the controlled-NOT. The matrix representation of the controlled-NOT, U_{CN}, is written with respect to the amplitudes for $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$, in that order.

Basic circuit example

Prepare superposition on one qubit: initialize $|0\rangle \xrightarrow{H} |+\rangle$.
Entangle two qubits: $|00\rangle \xrightarrow{H \text{ on qubit } 1} (|0\rangle + |1\rangle)|0\rangle/\sqrt{2} \xrightarrow{\text{CNOT}} (|00\rangle + |11\rangle)/\sqrt{2}$, which generates the Bell state.

1.2.4 Quantum teleportation



Protocol Overview

Quantum teleportation transfers an unknown single-qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ from Alice to Bob via pre-shared entanglement and two bits of classical communication, without physically moving the target qubit.

Key Steps

1. Pre-sharing: Prepare the Bell pair $|\Phi^+\rangle = \dfrac{|00\rangle + |11\rangle}{\sqrt{2}}$ between Alice and Bob.
2. Alice jointly measures her target qubit and her entangled qubit in the Bell basis.
3. Alice sends two classical bits of measurement result to Bob.
4. Bob applies corresponding Pauli corrections (I, X, Z, Y) according to the classical information, recovering the original state $|\psi\rangle$ on his side.

Let the unknown state be $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, with initial entangled pair $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. The total initial state is

$$|\psi_0\rangle = \frac{1}{\sqrt{2}}\left[\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|00\rangle + |11\rangle)\right].$$

After applying CNOT (control=qubit 1, target=qubit 2) and Hadamard on qubit 1, we obtain

$$|\psi_2\rangle = \frac{1}{2}\left[\alpha(|0\rangle + |1\rangle)(|00\rangle + |11\rangle) + \beta(|0\rangle - |1\rangle)(|10\rangle + |01\rangle)\right].$$

Regrouping terms by Alice's measurement basis gives

$$|\psi_2\rangle = \frac{1}{2}[|00\rangle\underbrace{(\alpha|0\rangle + \beta|1\rangle)}_{|\psi\rangle} + |01\rangle\underbrace{(\alpha|1\rangle + \beta|0\rangle)}_{X|\psi\rangle} + |10\rangle\underbrace{(\alpha|0\rangle - \beta|1\rangle)}_{Z|\psi\rangle} + |11\rangle\underbrace{(\alpha|1\rangle - \beta|0\rangle)}_{XZ|\psi\rangle}].$$

Alice measures her two qubits, sends the 2-bit result to Bob, who applies the corresponding Pauli correction $\{I, X, Z, ZX\}$ to recover $|\psi\rangle$.

1.2.5 Deutsch-Jozsa Algorithm

Problem statement Given a black-box oracle U_f implementing

$$U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle,$$

where $f : \{0,1\}^n \to \{0,1\}$ is guaranteed to be either *constant* (all $f(x) = 0$ or all $f(x) = 1$) or *balanced* (exactly half of inputs map to 0, half to 1). Determine whether f is constant with a single query to U_f.

Algorithm steps

1. Initialize: $|\psi_0\rangle = |0\rangle^{\otimes n}|1\rangle$
2. Apply Hadamard to all qubits:

$$|\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \cdot \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

3. Apply oracle U_f:

$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)}|x\rangle \cdot \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

4. Apply Hadamard to the first n qubits:

$$|\psi_3\rangle = \sum_{z=0}^{2^n-1} \left(\frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{f(x)+x\cdot z} \right) |z\rangle \cdot \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

5. Measure the first n qubits:
   - If f is constant: amplitude of $|0\rangle^{\otimes n}$ is 1, measurement result is $0^{\otimes n}$
   - If f is balanced: amplitude of $|0\rangle^{\otimes n}$ is 0, measurement result is non-zero

Key result The algorithm solves the problem with **1** query to U_f, achieving an exponential speedup over the classical worst-case requirement of $2^{n-1} + 1$ queries.

1.2.6 Classification and Theoretical Framework of Quantum Algorithms

Three fundamental categories construct the core system of quantum algorithms with provable computational advantages, with the **Hidden Subgroup Problem (HSP)** serving as the unified mathematical foundation for all exponential-speedup methods.

First, algorithms based on the quantum Fourier transform (QFT) achieve exponential acceleration. Represented by the Deutsch–Jozsa and Shors factoring/discrete logarithm algorithms, they rely on unitary discrete Fourier transforms acting upon qubit amplitudes. Defined over the group $\mathbb{Z}_{2^n}$, the standard QFT reads

$$\text{QFT} : |j\rangle \mapsto \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{\frac{2\pi i j k}{2^n}} |k\rangle,$$

reducing complexity from the classical FFT bound $O(N \log N)$ to $O(\log^2 N)$ for $N = 2^n$.

QFT enables efficient solutions to classically intractable problems including the Deutsch problem, integer factorization, and discrete logarithm. This paradigm inspired Kitaevs approach to abelian stabilizer problems, generalized into the canonical **Hidden Subgroup Problem**:

Hidden Subgroup Problem Given:

- a finitely generated group G,
- a finite set X,
- a function $f : G \to X$ constant on each coset of a subgroup $K \leq G$ and distinct across different cosets.

With a quantum oracle implementing

$$U|g\rangle|h\rangle = |g\rangle|h \oplus f(g)\rangle, \quad g \in G,\ h \in X,$$

where $\oplus$ denotes a specified binary operation on X, **recover a generating set of K.**

Second, Grover search provides quadratic speedup for unstructured exhaustive search via amplitude amplification with two-reflection interference. Using the operator $Q = -US_0U^\dagger S_\chi$, it reduces complexity from $O(N)$ to $O(\sqrt{N})$. This framework generalizes to combinatorial optimization and satisfiability problems, acting as a universal lightweight acceleration independent of HSP.

Third, quantum simulation resolves the classical intractability of many-body quantum systems. For a Hamiltonian $H = \sum_j H_j$, Trotterization approximates unitary evolution as

$$e^{-iHt} \approx \left(\prod_j e^{-iH_j t/M} \right)^M .$$

Quantum hardware encodes exponentially large Hilbert-space states with linearly scaled qubits; extracting physical observables remains limited by measurement collapse, with active research on Trotter error bounds and ground-state preparation via VQE/QPE.

In complexity theory, efficient quantum algorithms belong to the complexity class BQP (Bounded-Error Quantum Polynomial time), which represents problems efficiently solvable by a quantum computer with bounded error probability.

- P : (Polynomial time) is the class of problems efficiently solvable by a deterministic classical computer.
- NP :(Nondeterministic Polynomial time) consists of problems whose solutions can be verified efficiently in classical computation.
- PSPACE :denotes problems solvable using polynomial memory, regardless of time cost.

The fundamental inclusion relation is

$$\mathsf{P} \subseteq \mathsf{BQP} \subseteq \mathsf{PSPACE}.$$

It remains unproven whether BQP can efficiently solve NP-complete problems, the hardest problems in NP.

All quantum computational advantages essentially rely on superposition, interference and entanglement under unitary evolution constraints.

1.2.7 Fundamental Feasibility and Practical Implementation of Quantum Information Processing

This section discusses the core fundamental questions restricting large-scale quantum information processing, focusing on

(1) two critical concerns: **whether noise constitutes an insurmountable physical barrier**, and

(2)whether quantum mechanics remains complete for extended systems.

It clarifies that noise can be suppressed via quantum error correction and the threshold theorem as an **engineering solution** rather than a **theoretical limitation.**

Three hierarchical application scenarios are summarized:

(1) small-scale protocols including quantum tomography and

(2) quantum key distribution, medium-scale quantum simulation exceeding classical computational power, and

(3) large-scale cryptographic algorithms for factorization and discrete logarithm.

Three mainstream physical platformsphotonic systems, trapped atomic devices, and nuclear magnetic resonanceare analyzed, with hybrid architectures highlighted as the future direction for integrating technical advantages.

1.2.8 Classical Information over Quantum Channels

Classical information theory is founded on two fundamental theorems.

Shannon's Noiseless Channel Coding Theorem
This theorem determines the minimum number of bits required for storing outputs from an information source. It introduces the Shannon entropy:

$$H(p_j) \equiv -\sum_j p_j \log p_j$$

which sets a strict lower bound for lossless compression. The average code length per symbol cannot be smaller than $H(p_j)$; otherwise, high error rates will occur during decompression.

A classical information source is described by a probability distribution p_j and outputs symbols independently. A typical example is the character frequency in English texts, where natural redundancy can be exploited for compression.

This theorem satisfies the three core objectives of quantum information theory:

1. Identify fundamental static resources: classical bits and probabilistic information sources;
2. Identify fundamental dynamic processes: compression encoding and decompression recovery;
3. Quantify resource trade-offs: adopt Shannon entropy as the criterion for minimal compression resources.

Shannon's Noisy Channel Coding Theorem
This theorem quantifies the maximum reliable transmission rate over noisy channels. Redundancy is introduced to counteract channel noise. For instance, encoding one bit into two bits results in a channel capacity of approximately 0.5. The theorem provides a general method to calculate optimal redundancy and channel capacity for any fixed noise model.

This theorem also fulfills the three core objectives:

1. Identify fundamental static resources: original information sources and transmitted bits;
2. Identify fundamental dynamic processes: noise interference, error-correction encoding and decoding;
3. Quantify resource trade-offs: determine the optimal redundancy for reliable communication under a given noise model.

When classical information is transmitted via quantum states: noiseless quantum channels cannot improve classical compression efficiency. For noisy quantum channels, classical error correction is no longer applicable.

HSW Theorem (HolevoSchumacherWestmoreland)
This theorem provides a lower bound on the channel capacity for sending classical information through a noisy quantum channel. An open problem remains: whether capacity can exceed this bound without entanglement encoding. No conclusive proof exists to date.

1.2.9 Quantum Information over Quantum Channels

A quantum information source is defined by a probability distribution p_j together with corresponding quantum states $|\psi_j\rangle$, emitting pure states probabilistically.

Schumacher's Noiseless Quantum Channel Coding Theorem
Focusing on quantum state compression with fidelity approaching 1, the ultimate compression limit for non-orthogonal ensembles is given by the von Neumann entropy:

$$S(\rho) = -\operatorname{Tr}(\rho \log \rho)$$

Properties:

- For orthogonal states $|\psi_j\rangle$, $S(\rho) \equiv H(p_j)$, reducing to the classical Shannon limit;
- For non-orthogonal states such as $|0\rangle$ and $\frac{|0\rangle+|1\rangle}{\sqrt{2}}$, $S(\rho) < H(p_j)$, achieving better compression than classical schemes.

Example: a source generating $|0\rangle$ with probability p and the superposition state with probability $1-p$ can be compressed below the Shannon limit by utilizing inherent overlap redundancy in non-orthogonal quantum states.

This quantum coding theorem satisfies the three core objectives:

1. Identify fundamental static resources: quantum states described by density matrices, including superposition and entanglement resources;
2. Identify fundamental dynamic processes: quantum block compression, decoding, and fidelity control;
3. Quantify resource trade-offs: adopt von Neumann entropy as the threshold for optimal quantum storage and compression.

While noiseless quantum compression is well established, a complete coding theorem for general noisy quantum channels remains unfinished, with relevant progress discussed in subsequent chapters.

Quantum information theory represents the most abstract yet fundamental component within quantum computation and quantum information. The core questions driving this field and the entire discipline remain focused on three essential issues:

(1) What constitutes the essence of quantum information processing?

(2)What fundamental distinctions separate the quantum realm from the classical one?

(3)What unique resources unavailable in classical systems empower quantum computation?

Current understandings remain incomplete and ambiguous. Further research aims to clarify these topics, establishing a rigorous comprehension of both the potential and inherent limitations of quantum information processing.

Exercise 1.2: Identify the most significant discovery in quantum computation and quantum information to date, and write a 2000-word essay for an educated lay audience explaining this discovery.

1.3 Fundamental Postulates of Quantum Mechanics

We consider a complex separable Hilbert space $\mathcal{H}$ with inner product $\langle\cdot|\cdot\rangle$ and $\dim\mathcal{H} < \infty$.

1.3.1 Postulates for pure Quantum State

Postulate 1 ***(Pure State):*** *A pure state is a ray in $\mathcal{H}$:*

$$\underline{\psi} = \{\lambda|\psi\rangle \mid \lambda \in \mathbb{C}\}\,.$$

The pure state space is the projective Hilbert space:

$$\mathbb{P}(\mathcal{H}) = (\mathcal{H} \setminus \{0\})/\mathbb{C}^*.$$

A vector $|\psi\rangle \in \mathcal{H}$ is called a pure state vector. *We use a* normalized *vector with $\langle\psi|\psi\rangle = 1$ as its representative.*

Pure state: A point in the projective Hilbert space $\mathbb{P}(\mathcal{H})$.

Pure state vector: A vector $|\psi\rangle \in \mathcal{H}$ (representative of a state).

Postulates for Observables and measurement:

Postulate 2 ***(Observable):*** *Every observable corresponds to a self-adjoint operator $A = A^\dagger$. For discrete spectra:*

$$A = \sum_k \lambda_k \,|\lambda_k\rangle\langle\lambda_k|\,,$$

where $\lambda_k \in \mathbb{R}$ are eigenvalues (observed value).

Postulate 3 ***Measurement (Born Rule)*** *For a state $|\psi\rangle$, the probability to measure λ_k is*

$$p(\lambda_k) = \left|\langle\lambda_k|\,\psi\rangle\right|^2.$$

After measurement, the state collapses to $|\lambda_k\rangle$.

Postulates for evolution:

Postulate 4 ***(Schrödinger Equation):*** *The state of an isolated system satisfies*

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\,|\psi(t)\rangle = H\,|\psi(t)\rangle\,,$$

where $H = H^\dagger$ is the Hamiltonian.

Theorem (Unitary Evolution): If H is time-independent, then

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle\,, \quad U(t) = e^{-iHt/\hbar},$$

and $U(t)$ is unitary.

Postulates for composite Systems

Postulate 5 ***Composite Systems****: For any states of subsystems A and B,*

- *The composite state vector space is exactly spanned by all composite states $|\psi, \phi\rangle$ with $\psi \in \mathcal{H}_A$, $\phi \in \mathcal{H}_B$.*
- ***Statistical independence:*** *For any measurement outcomes a (on A) and b (on B), the joint probability factors as:*

$$\mathbb{P}(a \wedge b \mid |\psi, \phi\rangle) = \mathbb{P}(a \mid \psi)\, \mathbb{P}(b \mid \phi).$$

Lemma 4 ***(One-sided Inner Product Preservation):*** *For any fixed $|\phi\rangle \in \mathcal{H}_B$ and all $|\psi\rangle, |\psi'\rangle \in \mathcal{H}_A$,*

$$\langle \psi, \phi \mid \psi', \phi\rangle = \langle \psi \mid \psi'\rangle_A.$$

By symmetry, for any fixed $|\psi\rangle \in \mathcal{H}_A$ and all $|\phi\rangle, |\phi'\rangle \in \mathcal{H}_B$,

$$\langle \psi, \phi \mid \psi, \phi'\rangle = \langle \phi \mid \phi'\rangle_B.$$

Proof Fix an arbitrary $|\phi\rangle \in \mathcal{H}_B$. For any $|\psi\rangle, |\psi'\rangle \in \mathcal{H}_A$, consider the joint measurement of A in an orthonormal basis $\{|a\rangle\}$ and B in the state $|\phi\rangle$.

By the Born rule:

$$\mathbb{P}(a \wedge \phi \mid |\psi', \phi\rangle) = \left|\langle a, \phi \mid \psi', \phi\rangle\right|^2.$$

By the statistical independence postulate:

$$\mathbb{P}(a \wedge \phi \mid |\psi', \phi\rangle) = \mathbb{P}(a \mid \psi') \cdot \mathbb{P}(\phi \mid \phi) = \left|\langle a \mid \psi'\rangle\right|^2.$$

Equating the two expressions gives $|\langle a, \phi \mid \psi', \phi\rangle|^2 = |\langle a \mid \psi'\rangle|^2$ for all basis states $|a\rangle$. In finite dimensions, we can choose a global phase such that the inner products themselves are equal. The symmetric case follows identically. ■

Lemma 5 ***(Bilinearity of the Composite State Map):*** *The composite state map*

$$m : \mathcal{H}_A \times \mathcal{H}_B \to \mathcal{H}_{AB}, \quad m(|\psi\rangle, |\phi\rangle) = |\psi, \phi\rangle$$

is bilinear in both arguments.

Proof We first prove linearity in the first argument for any fixed $|\phi\rangle \in \mathcal{H}_B$. Define the vector explicitly dependent on ϕ:

$$v(\phi) = |\alpha\psi_1 + \beta\psi_2, \phi\rangle - \alpha|\psi_1, \phi\rangle - \beta|\psi_2, \phi\rangle \in \mathcal{H}_{AB}.$$

All terms in $v(\phi)$ share the same B-state ϕ, so $v(\phi) \in \mathcal{H}_{A,\phi} = \operatorname{span}\{|\psi, \phi\rangle \mid \psi \in \mathcal{H}_A\}$. By Hilbert space axioms, a vector orthogonal to all vectors in its own subspace is zero.

Take any $|\psi'\rangle \in \mathcal{H}_A$, so $|\psi', \phi\rangle \in \mathcal{H}_{A,\phi}$. Using Lemma 1:

$$\begin{aligned}
\langle \psi', \phi \mid v(\phi)\rangle &= \langle \psi', \phi \mid \alpha\psi_1 + \beta\psi_2, \phi\rangle - \alpha\langle \psi', \phi \mid \psi_1, \phi\rangle - \beta\langle \psi', \phi \mid \psi_2, \phi\rangle \\
&= \langle \psi' \mid \alpha\psi_1 + \beta\psi_2\rangle_A - \alpha\langle \psi' \mid \psi_1\rangle_A - \beta\langle \psi' \mid \psi_2\rangle_A \\
&= 0.
\end{aligned}$$

Thus $v(\phi) = 0$ for all $\phi \in \mathcal{H}_B$, so $m(\cdot, \phi)$ is linear. Linearity in the second argument follows symmetrically by defining $w(\psi) = |\psi, \alpha\phi_1 + \beta\phi_2\rangle - \alpha|\psi, \phi_1\rangle - \beta|\psi, \phi_2\rangle$ and repeating the argument. Hence m is bilinear. ■

Theorem 6 ***(Tensor Product of Composite Systems):*** *Let $\mathcal{H}_A$ and $\mathcal{H}_B$ be the state vector spaces of two quantum systems. Under the above postulate, the composite state vector space is isomorphic to*

$$\mathcal{H}_{AB} \simeq \mathcal{H}_A \otimes \mathcal{H}_B$$

as Hilbert spaces.

Proof By Lemma 2, the composite map $m : \mathcal{H}_A \times \mathcal{H}_B \to \mathcal{H}_{AB}$ is bilinear. By the composite system postulate, the image of m spans $\mathcal{H}_{AB}$.

The tensor product $\mathcal{H}_A \otimes \mathcal{H}_B$ is uniquely characterized (up to Hilbert space isomorphism) by the universal property: any bilinear map from $\mathcal{H}_A \times \mathcal{H}_B$ to a Hilbert space factors uniquely through the canonical bilinear map $\otimes : \mathcal{H}_A \times \mathcal{H}_B \to \mathcal{H}_A \otimes \mathcal{H}_B$.

Thus there exists a unique linear isomorphism $\tilde{m} : \mathcal{H}_A \otimes \mathcal{H}_B \to \mathcal{H}_{AB}$ such that

$$\tilde{m}(|\psi\rangle \otimes |\phi\rangle) = m(|\psi\rangle, |\phi\rangle) = |\psi, \phi\rangle$$

for all $|\psi\rangle \in \mathcal{H}_A$, $|\phi\rangle \in \mathcal{H}_B$. By Lemma 1 and bilinearity, $\tilde{m}$ preserves inner products:

$$\begin{aligned}\langle \tilde{m}(\psi' \otimes \phi') \mid \tilde{m}(\psi \otimes \phi)\rangle_{AB} &= \langle \psi', \phi' \mid \psi, \phi\rangle_{AB} \\ &= \langle \psi' \mid \psi\rangle_A \langle \phi' \mid \phi\rangle_B \\ &= \langle \psi' \otimes \phi' \mid \psi \otimes \phi\rangle_{\otimes}.\end{aligned}$$

Therefore $\tilde{m}$ is a Hilbert space isomorphism, and

$$\mathcal{H}_{AB} \simeq \mathcal{H}_A \otimes \mathcal{H}_B.$$

■

1.3.2 For Mixed Quantum States

Definition 7 (Density Operator) For a **statistical ensemble** of pure states $\{p_i, |\psi_i\rangle\}$ with $p_i \geq 0$ and $\sum_i p_i = 1$, the density operator is defined as

$$\rho = \sum_i p_i |\psi_i\rangle \langle\psi_i| .$$

The set of all such operators is denoted $\mathcal{D}(\mathcal{H})$.

Theorem 8 *The set of density operators can be characterized equivalently by*

$$\mathcal{D}(\mathcal{H}) = \{\rho \in \mathrm{End}(\mathcal{H}) \mid \rho^\dagger = \rho,\ \rho \succeq 0,\ \mathrm{tr}\,\rho = 1\}.$$

Proof ($\Rightarrow$) Let $\rho = \sum_i p_i |\psi_i\rangle \langle\psi_i|$ with $p_i \geq 0, \sum_i p_i = 1$.

- Self-adjoint: $\rho^\dagger = \sum_i p_i \big(|\psi_i\rangle \langle\psi_i| \big)^\dagger = \sum_i p_i |\psi_i\rangle \langle\psi_i| = \rho$.
- Positive semi-definite: For any $|\varphi\rangle \in \mathcal{H}$,

$$\langle\varphi|\rho|\varphi\rangle = \sum_i p_i \left|\langle\varphi|\psi_i\rangle\right|^2 \geq 0.$$

- Unit trace:

$$\mathrm{tr}(\rho) = \sum_i p_i \,\mathrm{tr}\left(|\psi_i\rangle \langle\psi_i| \right) = \sum_i p_i = 1.$$

($\Leftarrow$) Suppose $\rho^\dagger = \rho, \rho \succeq 0, \operatorname{tr}\rho = 1$. By spectral theorem, there exists an orthonormal eigenbasis $\{|\psi_i\rangle\}$ such that

$$\rho = \sum_i \lambda_i |\psi_i\rangle \langle\psi_i|, \quad \lambda_i \geq 0, \sum_i \lambda_i = \operatorname{tr}\rho = 1.$$

Set $p_i = \lambda_i$, then ρ is exactly the ensemble average of pure states $\{p_i, |\psi_i\rangle\}$. ■

Theorem 9 ***(Measurement for Mixed States)*** *For a density operator ρ and projection $P_k = |\lambda_k\rangle\langle\lambda_k|$, the measurement probability is*

$$p(\lambda_k) = \operatorname{tr}\big(\rho P_k\big).$$

After measurement, the state becomes

$$\rho_k = \frac{P_k \rho P_k}{p(\lambda_k)}.$$

Proof Let $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ be a statistical ensemble. By the Born rule for pure states, the probability of outcome k in state $|\psi_i\rangle$ is

$$p(k \mid \psi_i) = \langle\psi_i| P_k |\psi_i\rangle, \quad P_k = |\lambda_k\rangle\langle\lambda_k|.$$

Total probability gives

$$p(\lambda_k) = \sum_i p_i\, p(k \mid \psi_i) = \sum_i p_i \langle\psi_i| P_k |\psi_i\rangle.$$

Using $\operatorname{tr}(A|\psi\rangle\langle\psi|) = \langle\psi| A |\psi\rangle$, we rewrite

$$p(\lambda_k) = \sum_i p_i \operatorname{tr}\big(P_k |\psi_i\rangle\langle\psi_i|\big) = \operatorname{tr}\left(P_k \sum_i p_i |\psi_i\rangle\langle\psi_i|\right) = \operatorname{tr}\big(P_k\rho\big) = \operatorname{tr}\big(\rho P_k\big).$$

After measurement, each pure component collapses to $P_k|\psi_i\rangle$ normalized; averaging over the ensemble yields

$$\rho_k = \sum_i p_i \frac{P_k |\psi_i\rangle\langle\psi_i| P_k}{p(\lambda_k)} = \frac{P_k \left(\sum_i p_i |\psi_i\rangle\langle\psi_i|\right) P_k}{p(\lambda_k)} = \frac{P_k \rho P_k}{p(\lambda_k)}.$$

■

Theorem 10 ***(Unitary Evolution of Density Operators)*** *Under unitary evolution $U(t) = e^{-iHt/\hbar}$,*

$$\rho(t) = U(t)\rho(0)U(t)^\dagger,$$

which is equivalent to the von Neumann equation

$$i\hbar \frac{d}{dt}\rho(t) = \big[H, \rho(t)\big].$$

Proof Write the initial ensemble $\rho(0) = \sum_i p_i |\psi_i(0)\rangle\langle\psi_i(0)|$. From pure-state unitary evolution: $|\psi_i(t)\rangle = U(t)|\psi_i(0)\rangle$. Then

$$\begin{aligned}
\rho(t) &= \sum_i p_i |\psi_i(t)\rangle\langle\psi_i(t)| \\
&= \sum_i p_i\, U(t) |\psi_i(0)\rangle\langle\psi_i(0)| U(t)^\dagger \\
&= U(t) \left(\sum_i p_i |\psi_i(0)\rangle\langle\psi_i(0)|\right) U(t)^\dagger \\
&= U(t)\rho(0)U(t)^\dagger.
\end{aligned}$$

Differentiate $\rho(t) = U\rho U^\dagger$ with $U = e^{-iHt/\hbar}$, $\dot{U} = -\frac{i}{\hbar}HU$, $\dot{U}^\dagger = \frac{i}{\hbar}U^\dagger H$:

$$\dot{\rho} = \dot{U}\rho U^\dagger + U\rho\dot{U}^\dagger = -\tfrac{i}{\hbar}HU\rho U^\dagger + \tfrac{i}{\hbar}U\rho U^\dagger H = \tfrac{i}{\hbar}\big(\rho H - H\rho\big).$$

Rearranged:

$$i\hbar\dot{\rho} = [H, \rho].$$

■

Definition 11 (Partial Trace & Reduced Density Matrix) The partial trace over subsystem B is the linear operator

$$\mathrm{tr}_B\colon \quad \mathcal{L}(\mathcal{H}_A \otimes \mathcal{H}_B) \to \mathcal{L}(\mathcal{H}_A)$$

uniquely defined by the identity

$$\mathrm{tr}_B = \mathrm{id}_A \otimes \mathrm{tr}_B,$$

For a composite state ρ_{AB}, the *reduced density matrix* of subsystem A is

$$\rho_A = \mathrm{tr}_B(\rho_{AB}).$$

Act on product operators as

$$\mathrm{tr}_B\big(A \otimes B\big) = A \cdot \mathrm{tr}(B).$$

Theorem 12 *Let $\mathcal{H}_A = \mathrm{span}\{|i\rangle_A\}$ and $\mathcal{H}_B = \mathrm{span}\{|\mu\rangle_B\}$. For any operator $\rho \in \mathcal{L}(\mathcal{H}_A \otimes \mathcal{H}_B)$, the partial trace $\mathrm{tr}_B = \mathrm{id}_A \otimes \mathrm{tr}_B$ gives*

$$\langle i|_A \,\mathrm{tr}_B(\rho)\, |j\rangle_A = \sum_\mu \langle i, \mu|\, \rho\, |j, \mu\rangle\,.$$

Proof By linearity, expand ρ in the product basis:

$$\rho = \sum_{i,j,\mu,\nu} \rho_{i\mu,j\nu}\, |i\mu\rangle\langle j\nu|\,.$$

Apply the definition $\mathrm{tr}_B(A \otimes B) = A \cdot \mathrm{tr}(B)$:

$$\mathrm{tr}_B\big(\,|i\rangle\langle j| \otimes |\mu\rangle\langle\nu|\,\big) = |i\rangle\langle j| \cdot \mathrm{tr}\big(\,|\mu\rangle\langle\nu|\,\big) = |i\rangle\langle j| \cdot \delta_{\mu\nu}.$$

Thus

$$\mathrm{tr}_B(\rho) = \sum_{i,j,\mu} \rho_{i\mu,j\mu}\, |i\rangle\langle j|\,.$$

Taking matrix elements on $\mathcal{H}_A$ yields

$$\langle i|\,\mathrm{tr}_B(\rho)\,|j\rangle = \sum_\mu \langle i\mu|\,\rho\,|j\mu\rangle\,.$$

■

Theorem 13 ***(Pure State Criterion)*** *A density operator ρ represents a pure state if and only if*

$$\rho^2 = \rho \quad \Longleftrightarrow \quad \mathrm{tr}(\rho^2) = 1.$$

In this case, $\rho = |\psi\rangle\,\langle\psi|$ for some normalized state vector $|\psi\rangle$.

Proof We prove both directions of the equivalence.

(⇒) Suppose $\rho = |\psi\rangle\langle\psi|$ is a pure state with $\langle\psi|\psi\rangle = 1$. Then

$$\rho^2 = \big(|\psi\rangle\langle\psi|\big)\big(|\psi\rangle\langle\psi|\big) = |\psi\rangle\big(\langle\psi|\psi\rangle\big)\langle\psi| = |\psi\rangle\langle\psi| = \rho.$$

Taking the trace:

$$\mathrm{tr}(\rho^2) = \mathrm{tr}(\rho) = 1.$$

(⇐) Suppose ρ is a density operator satisfying $\rho^2 = \rho$. By the spectral theorem,

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|, \quad p_k \geq 0, \ \sum_k p_k = 1.$$

Then

$$\rho^2 = \sum_k p_k^2 |\psi_k\rangle\langle\psi_k|.$$

Since $\rho^2 = \rho$, we have $p_k^2 = p_k$ for all k, so $p_k \in \{0, 1\}$. Because $\sum_k p_k = 1$, exactly one $p_k = 1$ and all others vanish. Thus $\rho = |\psi\rangle\langle\psi|$ for some normalized $|\psi\rangle$.

Finally, if $\mathrm{tr}(\rho^2) = 1$, then $\sum_k p_k^2 = 1$ together with $\sum_k p_k = 1$ and $p_k \geq 0$ implies exactly one $p_k = 1$, so ρ is pure. ■

Summary

1. **Pure States**: Equivalence classes (rays) in $\mathcal{H}$, represented by normalized $|\psi\rangle \in \mathbb{P}(\mathcal{H})$.
2. **Observables & Measurement**: Self-adjoint operators; Born rule $p(\lambda_k) = |\langle\lambda_k|\psi\rangle|^2$ with state collapse.
3. **Dynamics**: Schrödinger equation yields unitary evolution $U = e^{-iHt/\hbar}$.
4. **Composite Systems**: $\mathcal{H}_{AB} \simeq \mathcal{H}_A \otimes \mathcal{H}_B$ via statistical independence and bilinearity.
5. **Density Operators**: Mixed states satisfy $\rho^\dagger = \rho,\ \rho \succeq 0,\ \mathrm{tr}\,\rho = 1$.
6. **Mixed-state Rules**: $p(\lambda_k) = \mathrm{tr}(\rho P_k)$; $\rho(t) = U\rho U^\dagger$ obeys the von Neumann equation.
7. **Partial Trace**: $\rho_A = \mathrm{tr}_B(\rho_{AB})$ characterizes subsystem states.
8. **Pure Criterion**: $\rho^2 = \rho \iff \mathrm{tr}(\rho^2) = 1$.

1.4 Plan for Next Week

Focus on the subsequent two chapters of Nielsen & Chuang:

1. Study standard advanced content, sort frameworks and supplement detailed proofs for textbook derivations.
2. Integrate new knowledge into the existing note system with unified logic and formatting.
3. Add personal comprehension during reading, and polish the overall coherence of the full notes.

1.5 Appendix

The appendix provides a concise summary of my theoretical progress, including successful outcomes, failures, or ongoing experimentation conducted throughout the week.

PS: All technical/experimental details pertaining to the theorem will be or have been summarized in the specified notes.

1.5.1 Theorem Proofs

Theorem 14 *The matrix exponential map*

$$\exp : \mathbb{C}^{n\times n} \longrightarrow \mathrm{GL}_n(\mathbb{C}), \quad X \longmapsto e^X$$

is surjective.

Proof We use the **Jordan-Chevalley decomposition** for complex matrices.

Lemma 15 ***(Jordan-Chevalley Decomposition):****any matrix $A \in \mathbb{C}^{n\times n}$ admits a unique decomposition*

$$A = S + N,$$

where S is diagonalizable, N is nilpotent, and $SN = NS$.

Let $A \in \mathrm{GL}_n(\mathbb{C})$ be arbitrary. We construct $X \in \mathbb{C}^{n\times n}$ such that $e^X = A$ in four steps:

Step 1: Multiplicative decomposition of A

Since A is invertible, its diagonalizable part S must also be invertible (otherwise S would have a zero eigenvalue, implying A has a zero eigenvalue, contradicting invertibility).

Define $M = S^{-1}N$. Since $SN = NS$, we have $SM = MS$, and M is nilpotent (because $N^k = 0$ implies $M^k = S^{-k}N^k = 0$). Thus:

$$A = S(I + M), \quad \text{and } S \text{ commutes with } I + M.$$

Step 2: Logarithm of the diagonalizable part S

Since S is diagonalizable and invertible, there exists an invertible matrix P such that

$$S = PDP^{-1}, \quad D = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n), \quad \lambda_i \neq 0.$$

For each non-zero complex number λ_i, choose any complex logarithm $\ln \lambda_i$ (which always exists in $\mathbb{C}$). Define:

$$L_S = P \cdot \mathrm{diag}(\ln \lambda_1, \ln \lambda_2, \ldots, \ln \lambda_n) \cdot P^{-1}.$$

By the properties of matrix exponentials:

$$e^{L_S} = P \cdot \mathrm{diag}(e^{\ln \lambda_1}, \ldots, e^{\ln \lambda_n}) \cdot P^{-1} = PDP^{-1} = S.$$

Step 3: Logarithm of the unipotent part $I + M$

Since M is nilpotent, $M^n = 0$. We define the logarithm of $I + M$ via the finite power series:

$$L_M = \ln(I + M) = \sum_{k=1}^{n-1} \frac{(-1)^{k+1}}{k} M^k.$$

This series terminates after $n-1$ terms and converges absolutely. A standard power series identity shows that:

$$e^{L_M} = I + M.$$

Step 4: Combine the logarithms

Since S commutes with M, L_S (a polynomial in S) commutes with L_M (a polynomial in M). For commuting matrices, $e^{X+Y} = e^X e^Y$, so:

$$e^{L_S+L_M} = e^{L_S}e^{L_M} = S(I+M) = A.$$

Let $X = L_S + L_M \in \mathbb{C}^{n\times n}$. Then $e^X = A$. Since $A \in \mathrm{GL}_n(\mathbb{C})$ was arbitrary, the exponential map is surjective. ■

Theorem 16 *The matrix exponential map*

$$\exp : \mathfrak{su}_n(\mathbb{C}) \longrightarrow \mathrm{SU}_n(\mathbb{C}), \quad X \longmapsto e^X$$

is surjective.

Proof Recall the definitions:

- $\mathfrak{su}_n(\mathbb{C}) = \{X \in \mathbb{C}^{n\times n} \mid X^\dagger = -X, \mathrm{tr}(X) = 0\}$ (trace-zero anti-Hermitian matrices)
- $\mathrm{SU}_n(\mathbb{C}) = \{U \in \mathbb{C}^{n\times n} \mid U^\dagger U = I, \det(U) = 1\}$ (unitary matrices with determinant 1)

Let $U \in \mathrm{SU}_n(\mathbb{C})$ be arbitrary. Since U is unitary, it is a normal matrix. By the **spectral theorem for normal matrices**, there exists a unitary matrix P such that

$$U = PDP^\dagger,$$

where $D = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ is a diagonal matrix with $|\lambda_k| = 1$ for all k.

The determinant condition gives:

$$\det(U) = \det(D) = \prod_{k=1}^{n} \lambda_k = 1.$$

For each unit-modulus complex number λ_k, there exists a real number θ_k such that $\lambda_k = e^{i\theta_k}$. Substituting into the determinant condition:

$$\prod_{k=1}^{n} e^{i\theta_k} = e^{i(\theta_1+\theta_2+\cdots+\theta_n)} = 1,$$

which implies

$$\theta_1 + \theta_2 + \cdots + \theta_n = 2\pi m$$

for some integer $m \in \mathbb{Z}$.

Using the multi-valuedness of the complex logarithm, we choose integers $k_1, k_2, \ldots, k_n$ such that

$$\sum_{j=1}^{n} (\theta_j + 2\pi k_j) = 0.$$

For example, we can take $k_1 = -m$ and $k_2 = k_3 = \cdots = k_n = 0$, which satisfies the condition.

Now define the matrix:

$$L = P \cdot \mathrm{diag}\left(i(\theta_1 + 2\pi k_1), i(\theta_2 + 2\pi k_2), \ldots, i(\theta_n + 2\pi k_n)\right) \cdot P^\dagger.$$

We verify that $L \in \mathfrak{su}_n(\mathbb{C})$:

1. **Anti-Hermiticity**:
$$L^\dagger = P \cdot \operatorname{diag}\left(-i(\theta_1 + 2\pi k_1), \ldots, -i(\theta_n + 2\pi k_n)\right) \cdot P^\dagger = -L.$$
2. **Trace zero**:
$$\operatorname{tr}(L) = \operatorname{tr}\left(\operatorname{diag}\left(i(\theta_1 + 2\pi k_1), \ldots, i(\theta_n + 2\pi k_n)\right)\right) = i\sum_{j=1}^{n}(\theta_j + 2\pi k_j) = 0.$$

Finally, we compute the exponential of L:
$$\begin{aligned} e^L &= P \cdot \operatorname{diag}\left(e^{i(\theta_1+2\pi k_1)}, \ldots, e^{i(\theta_n+2\pi k_n)}\right) \cdot P^\dagger \\ &= P \cdot \operatorname{diag}\left(e^{i\theta_1}, \ldots, e^{i\theta_n}\right) \cdot P^\dagger \\ &= PDP^\dagger = U. \end{aligned}$$

Since $U \in \mathrm{SU}_n(\mathbb{C})$ was arbitrary, the exponential map is surjective. ■

Partition Induced by the Surjective Exponential Map

The surjective map $\exp : \mathfrak{su}_n(\mathbb{C}) \to \mathrm{SU}_n(\mathbb{C})$ induces a partition of $\mathfrak{su}_n$ into disjoint fibers:
$$\mathfrak{su}_n = \bigsqcup_{U \in \mathrm{SU}_n} F_U, \quad F_U = \{X \in \mathfrak{su}_n \mid e^X = U\}.$$

Theorem 17 *Let $U \in \mathrm{SU}_n(\mathbb{C})$ with spectral decomposition*
$$U = P \cdot \operatorname{diag}(e^{i\theta_1}, e^{i\theta_2}, \ldots, e^{i\theta_n}) \cdot P^\dagger, \quad \theta_k \in (-\pi, \pi].$$
*Then **all solutions** $X \in \mathfrak{su}_n(\mathbb{C})$ to $e^X = U$ are given by*
$$X = P \cdot \operatorname{diag}\left(i(\theta_1 + 2\pi k_1), i(\theta_2 + 2\pi k_2), \ldots, i(\theta_n + 2\pi k_n)\right) \cdot P^\dagger,$$
where $k_1, k_2, \ldots, k_n \in \mathbb{Z}$ are integers satisfying the trace-zero condition
$$\sum_{j=1}^{n} k_j = -\frac{1}{2\pi}\sum_{j=1}^{n}\theta_j = -m, \quad m \in \mathbb{Z}.$$

Proof ($\Rightarrow$) Suppose $e^X = U$. Since U is unitary (hence normal), X must commute with U. Therefore, X is simultaneously diagonalizable with U by the same unitary matrix P:
$$X = P \cdot \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_n) \cdot P^\dagger.$$
The condition $e^X = U$ implies $e^{\mu_j} = e^{i\theta_j}$ for all j, so $\mu_j = i(\theta_j + 2\pi k_j)$ for some integers $k_j \in \mathbb{Z}$.

Since $X \in \mathfrak{su}_n(\mathbb{C})$, it must be trace-zero:
$$\operatorname{tr}(X) = \sum_{j=1}^{n}\mu_j = i\sum_{j=1}^{n}(\theta_j + 2\pi k_j) = 0.$$
This gives the integer condition
$$\sum_{j=1}^{n} k_j = -\frac{1}{2\pi}\sum_{j=1}^{n}\theta_j = -m.$$

($\Leftarrow$) Conversely, any matrix of this form is anti-Hermitian and trace-zero (hence in $\mathfrak{su}_n$), and a direct computation shows $e^X = U$. ■

1.5.2 Discussion

This section clarifies the status of the partition induced by $\exp : \mathfrak{su}_n(\mathbb{C}) \to \mathrm{SU}_n(\mathbb{C})$, a prototypical example of a compact **semisimple Lie group**.

Classical, Fully Resolved Results (Not Open Problems)

All core results are standard textbook material for compact semisimple Lie groups:

- **Surjectivity**: A fundamental theorem states that the exponential map of any compact connected Lie group is surjective.
- **Fiber structure**: For compact semisimple Lie groups, each fiber $F_U = \{X \mid e^X = U\}$ is a discrete affine lattice, a translate of the coroot lattice kernel $\ker(\exp) = 2\pi\Lambda$. For $\mathfrak{su}_n$, this kernel has the explicit form:

$$\ker(\exp) = \left\{ 2\pi i \cdot \mathrm{diag}(k_1, k_2, \ldots, k_n) \mid k_j \in \mathbb{Z}, \sum_{j=1}^{n} k_j = 0 \right\}.$$

- **Representatives**: Each fiber contains a unique principal representative (via the principal logarithm branch $(-\pi, \pi]$) and a unique minimal-norm element (corresponding to the shortest geodesic in the bi-invariant metric).
- **Low-dimensional topology**: For $\mathfrak{su}_2$, the fundamental domain is a 3-ball of radius π with antipodal points identified, realizing $\mathrm{SU}_2 \cong S^3$.

Active Open Research Directions

While the pure mathematical foundations are closed, important applied questions remain, particularly for quantum information:

- **Optimal fundamental domains**: No consensus on the "best" domain for quantum gate decomposition (balancing volume, symmetry, smoothness, and tensor compatibility).
- **Smooth inverses**: Existence of global Lipschitz or smooth inverses (except on a low-dimensional set) for quantum optimal control.
- **Numerical stability**: Stable algorithms for large $n = 2^k$ (multi-qubit systems) near the branch cut.
- **Composite systems**: Interaction with tensor product structures for scalable Trotterization and gate decomposition.

In summary, the theory for compact semisimple Lie groups (including SU_n) is fully classical. In contrast, exponential map theory for non-compact semisimple Lie groups remains largely open, and quantum applications have created new practical research directions.

References

[1] G. Carcassi, L. Maccone, and C. A. Aidala. The four postulates of quantum mechanics are three. *Physical Review Letters*, 126(11):110402, 2021. arXiv:2003.11007.

[2] M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information*, 10th Anniversary Edition. Cambridge University Press, 2010.

[3] M. Hayashi. *Group Representation for Quantum Theory*. Springer, 2017.